EXP 2 : MODELING OF HEXAGONAL CELLULAR COVERAGE: ADJACENT CELLS, CELL SPLITTING, USER CAPACITY AND HANDOFF

MATLAB CODE:

```
clc;
clear;
close all;
%% HEXAGONAL CELLULAR COVERAGE SIMULATION
R = 1;                % Cell radius
usersPerCell = 50;    % Maximum users per cell
numRings = 2;         % Number of adjacent cell rings
%% 1. GENERATE HEXAGONAL CELL LOCATIONS
centers = [];
for q = -numRings:numRings
    for r = -numRings:numRings
        if abs(q+r) <= numRings
            x = 1.5*R*q;
            y = sqrt(3)*R*(r+q/2);
            centers = [centers; x y];
        end
    end
end
numCells = size(centers,1);
%% 2. HEXAGONAL CELLULAR COVERAGE
figure('Name','Hexagonal Cellular Coverage');
hold on;
axis equal;
grid on;
for i = 1:numCells
```

```
    x = centers(i,1);
    y = centers(i,2);
    drawHex(x,y,R,'cyan');
    text(x,y,num2str(i), ...
        'HorizontalAlignment','center', ...
        'FontWeight','bold');
end
title('Hexagonal Cellular Network Coverage');
xlabel('X Position');
ylabel('Y Position');
hold off;
fprintf('\n--- CELLULAR COVERAGE ---\n');
fprintf('Total Number of Cells: %d\n',numCells);
%% 3. ADJACENT CELL REPRESENTATION
figure('Name','Adjacent Cells');
hold on;
axis equal;
grid on;
for i = 1:numCells
    drawHex(centers(i,1),centers(i,2),R,'yellow');
end
plot(centers(:,1),centers(:,2),'r.','MarkerSize',15);
title('Adjacent Hexagonal Cells');
xlabel('X Position');
ylabel('Y Position');
hold off;
%% 4. CELL SPLITTING
figure('Name','Cell Splitting');
hold on;
axis equal;
grid on;
% Original large cell
```

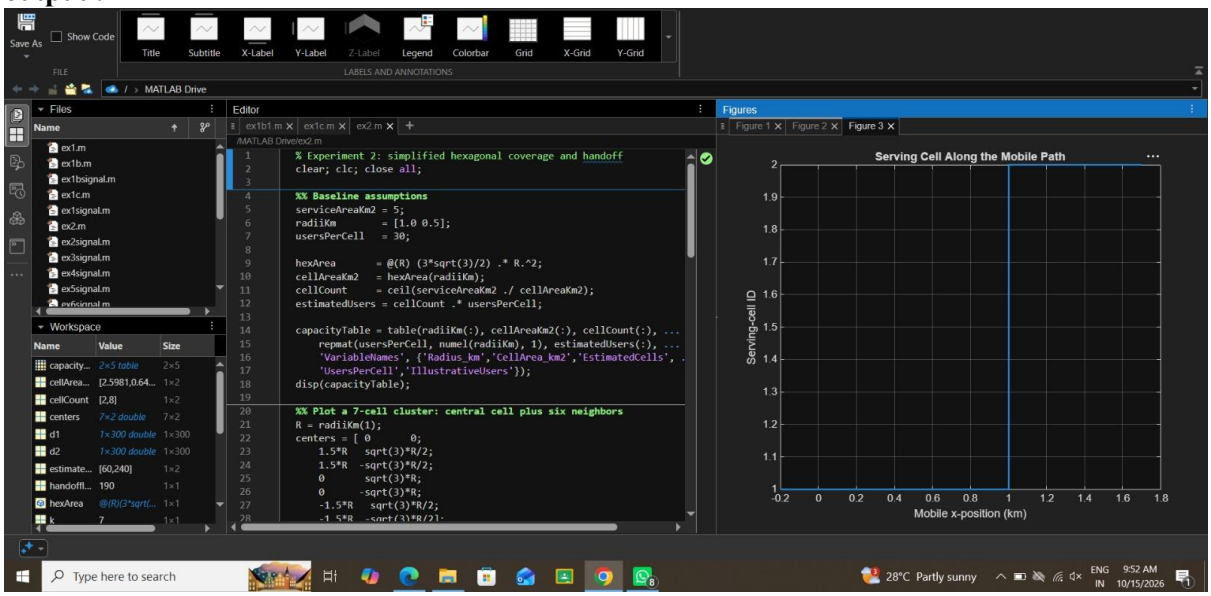
```
drawHex(0,0,R,'cyan');
% Divide the cell into seven smaller cells
smallR = R/3;
drawHex(0,0,smallR,'red');
for k = 0:5
    angle = k*pi/3;
    x = sqrt(3)*smallR*cos(angle);
    y = sqrt(3)*smallR*sin(angle);
    drawHex(x,y,smallR,'green');
end
title('Cell Splitting: One Large Cell into Seven Small Cells');
xlabel('X Position');
ylabel('Y Position');
legend('Original Cell','Small Cells');
hold off;
%% 5. USER CAPACITY CALCULATION
initialCapacity = numCells*usersPerCell;
% One cell split into seven smaller cells
newCapacity = (numCells-1)*usersPerCell + ...
            7*usersPerCell;
capacityIncrease = newCapacity-initialCapacity;
fprintf('\n--- USER CAPACITY ---\n');
fprintf('Number of Cells: %d\n',numCells);
fprintf('Users Per Cell: %d\n',usersPerCell);
fprintf('Initial Network Capacity: %d Users\n', ...
    initialCapacity);
fprintf('Capacity After Cell Splitting: %d Users\n', ...
    newCapacity);
fprintf('Additional Users Supported: %d\n', ...
    capacityIncrease);
%% 6. RANDOM USER DISTRIBUTION
figure('Name','User Distribution');
```

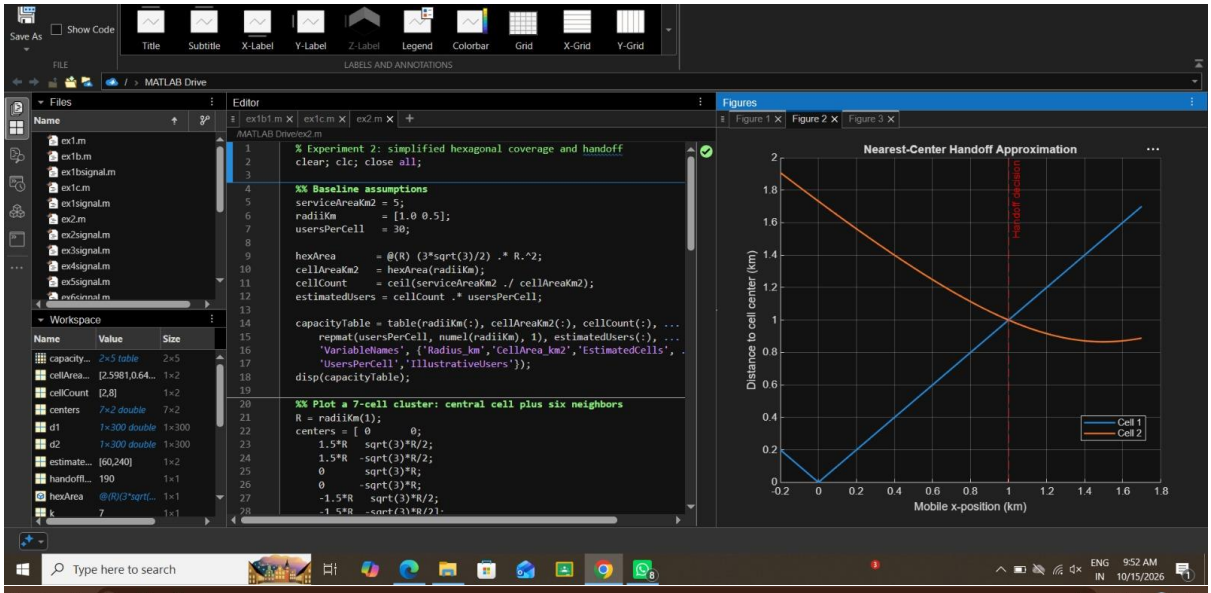
```
hold on;
axis equal;
grid on;
for i = 1:numCells
    drawHex(centers(i,1),centers(i,2),R,'cyan');
end
% Generate random users in the network area
numUsers = 100;
userX = [];
userY = [];
while length(userX) < numUsers
    x = (rand-0.5)*6*R;
    y = (rand-0.5)*6*R;
    % Check whether point belongs to the network
    distances = sqrt(sum((centers-[x y]).^2,2));
    if min(distances) < 0.85*R
        userX = [userX x];
        userY = [userY y];
    end
end
scatter(userX,userY,15,'r','filled');
title('Random Mobile User Distribution');
xlabel('X Position');
ylabel('Y Position');
legend('Cell Coverage','Mobile Users');
hold off;
%% 7. HANDOFF SIMULATION
figure('Name','Mobile Handoff');
hold on;
axis equal;
grid on;
```

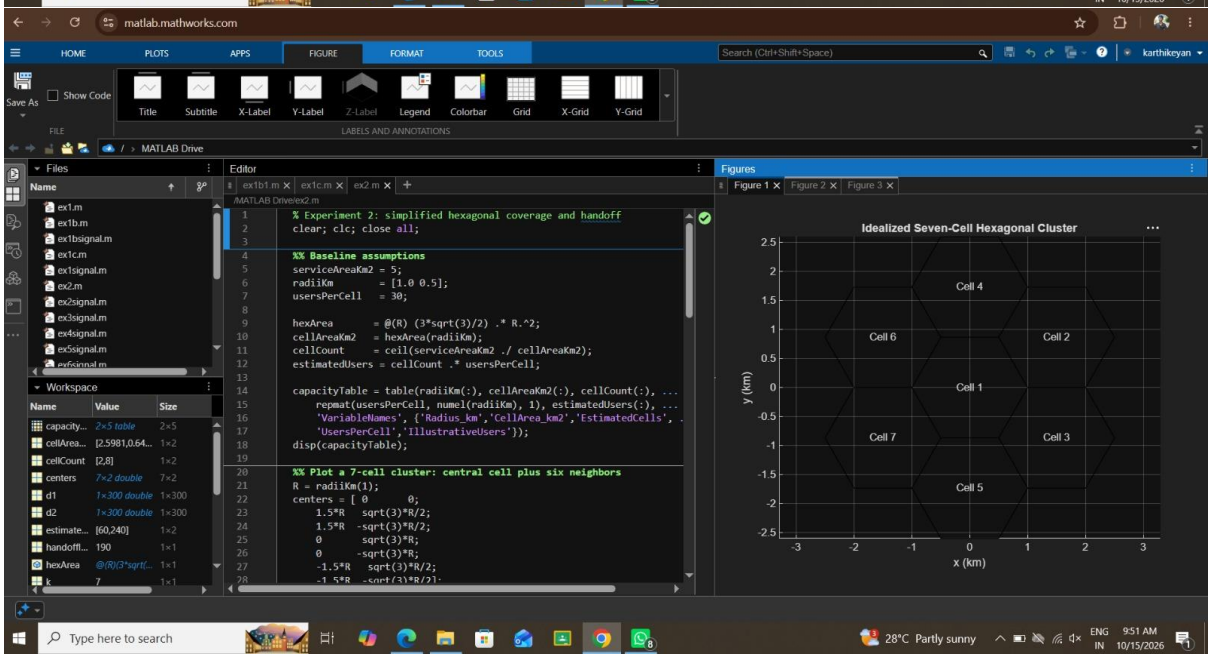
```
% Draw three adjacent cells
handoffCenters = [0 0;
                  1.5*R 0;
                  3*R 0];
for i = 1:3
    drawHex(handoffCenters(i,1), ...
            handoffCenters(i,2),R,'cyan');
    text(handoffCenters(i,1), ...
         handoffCenters(i,2), ...
         ['Cell ' num2str(i)], ...
         'HorizontalAlignment','center');
end
% Mobile user movement
pathX = linspace(-0.5,3.5,100);
pathY = 0.15*sin(linspace(0,2*pi,100));
plot(pathX,pathY,'r--','LineWidth',2);
% Show mobile user positions
for k = 1:10:100
    plot(pathX(k),pathY(k),'ro', ...
        'MarkerFaceColor','red','MarkerSize',8);
end
title('Mobile User Movement and Handoff');
xlabel('X Position');
ylabel('Y Position');
legend('Cell Coverage','Mobile User Path','Mobile User');
hold off;
%% 8. HANDOFF DETECTION
fprintf('\n--- HANDOFF SIMULATION ---\n');
previousCell = 1;
for k = 1:100
    userPosition = [pathX(k),pathY(k)];
```

```
    distances = sqrt(sum( ...
        (handoffCenters-userPosition).^2,2));
    [~,currentCell] = min(distances);
    if currentCell ~= previousCell
        fprintf('Handoff: Cell %d -> Cell %d\n', ...
            previousCell,currentCell);
        previousCell = currentCell;
    end
end
%% 9. FINAL RESULTS
fprintf('\n--- FINAL SIMULATION RESULTS ---\n');
fprintf('Total Hexagonal Cells: %d\n',numCells);
fprintf('Initial Capacity: %d Users\n',initialCapacity);
fprintf('Capacity After Splitting: %d Users\n',newCapacity);
fprintf('Capacity Improvement: %.2f %%\n', ...
    (capacityIncrease/initialCapacity)*100);
fprintf('\nSimulation Completed Successfully!\n');
%% HEXAGON DRAWING FUNCTION
function drawHex(x,y,R,colorName)
    theta = (0:6)*pi/3;
    hx = x+R*cos(theta);
    hy = y+R*sin(theta);
    patch(hx,hy,colorName, ...
        'FaceAlpha',0.25, ...
        'EdgeColor','blue', ...
        'LineWidth',1.5);
end
```

output :


Save As
Show Code
FILE
Title
Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
LABELS AND ANNOTATIONS
MATLAB Drive
Files
Name
ex1.m
ex1b.m
ex1bsignal.m
ex1c.m
ex1signal.m
ex2.m
ex2signal.m
ex3signal.m
ex4signal.m
ex5signal.m
Workspace
Name
Value
Size
capacity... 2×5 table 2×5
cellArea... [2.5981,0.64... 1×2
cellCount [2,8] 1×2
centers 7×2 double 7×2
d1 1×300 double 1×300
d2 1×300 double 1×300
estimate... [60,240] 1×2
handoffl... 190 1×1
hexArea @(R)(3*sqrt(... 1×1
k 7 1×1
Editor
ex1b1.m
ex1c.m
ex2.m
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% Experiment 2: simplified hexagonal coverage and handoff
clear; clc; close all;
%% Baseline assumptions
serviceAreaKm2 = 5;
radiiKm = [1.0 0.5];
usersPerCell = 30;
hexArea = @(R) (3*sqrt(3)/2) .* R.^2;
cellAreaKm2 = hexArea(radiiKm);
cellCount = ceil(serviceAreaKm2 ./ cellAreaKm2);
estimatedUsers = cellCount .* usersPerCell;
capacityTable = table(radiiKm(:), cellAreaKm2(:), cellCount(:), ...
repmat(usersPerCell, numel(radiiKm), 1), estimatedUsers(:), ...
'VariableNames', {'Radius_km','CellArea_km2','EstimatedCells', ...
'UsersPerCell','IllustrativeUsers'});
disp(capacityTable);
%% Plot a 7-cell cluster: central cell plus six neighbors
R = radiiKm(1);
centers = [0 0;
1.5*R sqrt(3)*R/2;
1.5*R -sqrt(3)*R/2;
0 sqrt(3)*R;
0 -sqrt(3)*R;
-1.5*R sqrt(3)*R/2;
-1.5*R -sqrt(3)*R/2];
Figures
Figure 1
Figure 2
Figure 3
Serving Cell Along the Mobile Path
Serving-cell ID
Mobile x-position (km)
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28°C Partly sunny
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ex4signal.m
ex5signal.m
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cellCount
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d2
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1.5*R -sqrt(3)*R/2;
0 sqrt(3)*R;
0 -sqrt(3)*R;
-1.5*R sqrt(3)*R/2;
-1.5*R -sqrt(3)*R/2];
Figures
Figure 1
Figure 2
Figure 3
Nearest-Center Handoff Approximation
Handoff decision
Distance to cell center (km)
Mobile x-position (km)
Cell 1
Cell 2
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1.5*R -sqrt(3)*R/2;
0 sqrt(3)*R;
0 -sqrt(3)*R;
-1.5*R sqrt(3)*R/2;
-1.5*R -sqrt(3)*R/2];
Figures
Figure 1
Figure 2
Figure 3
Idealized Seven-Cell Hexagonal Cluster
Cell 4
Cell 6
Cell 2
Cell 1
Cell 7
Cell 3
Cell 5
y (km)
x (km)
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28°C Partly sunny
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